Week 13: Target Trial Emulation

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Overview

Welcome, class, to STAT 579. We begin our journey into causal inference with what I consider the single most important conceptual tool for applied research: the **Target Trial** framework (Hernán et al., 2016).

In your statistical training so far, you have become experts at quantifying *associations*. You can build complex models to predict Y from A. This course, however, is about a fundamentally different question: "What is the **causal effect** of A on Y?"

The central idea of the target trial framework is simple: every observational analysis that aims to estimate a causal effect is, implicitly, an *emulation* of a hypothetical, pragmatic randomized trial. This hypothetical, ideal experiment is what we call the "target trial."

Why start here? Because this framework is the foundational *thinking process* that must precede any statistical analysis. Before you run a single regression or calculate a single weight, you must first design the experiment you *wish* you could have run.

By explicitly designing this target trial *before* you analyze your observational data, you create a clear blueprint. This blueprint serves two critical purposes:

1. **It clarifies your causal question.** A vague question like "what is the effect of aspirin?" becomes a precise, answerable question: "What is the 5-year risk of death if individuals with colon cancer *initiate* daily aspirin 1-month post-surgery, compared with the 5-year risk if they do not initiate aspirin?"
2. **It protects you from common, severe biases.** As we will see, many infamous biases in observational research, such as **immortal time bias**, are "self-inflicted injuries" (Hernán et al. (2016)). They are not inherent to the data but are created by a flawed analysis plan. Explicitly emulating a target trial protocol prevents these errors.

In this lecture, we will define the seven key components of a target trial protocol, distinguish between the two key causal questions (ITT vs. PP), and use this framework to design and critique observational studies.

Learning Objectives

By the end of this session, you will be able to:

1. **Define** the "target trial" framework and articulate its role as a unifying structure for designing and critiquing observational research.
2. **Identify** and **explain** the seven essential components of a target trial protocol (Eligibility Criteria, Treatment Strategies, Assignment Procedure, Follow-up Period, Outcome, Causal Contrast, and Analysis Plan).
3. **Differentiate** between the **Intention-to-Treat (ITT) effect** and the **Per-Protocol (PP) effect** as two distinct and valid causal contrasts, and explain the different causal questions they answer.
4. **Explain** *why* a "naïve" per-protocol analysis (e.g., comparing only the adherers in a trial) is biased, using causal diagrams and the concept of post-randomization confounding.
5. **Recognize** and **diagnose** common and severe emulation failures related to `Time Zero` misalignment, including **immortal time bias** and **prevalent user bias**.
6. **Propose** valid emulation strategies (e.g., new-user design, grace periods) to address common challenges in observational data.

R Packages

Today's lecture is primarily conceptual, but we will use the following packages to visualize our causal assumptions and prepare data. The lab will use `tidyverse` and `survival`.

```r
# Load the pacman package, installing if-needed
if (!require("pacman")) {
  install.packages("pacman")
}

# Load (and install if-needed) packages for this session
pacman::p_load(
  tidyverse, # For all data wrangling and plotting
  ggdag, # For plotting DAGs with ggplot2
  dagitty, # For creating and analyzing DAGs
  knitr, # For table formatting
  kableExtra, # For prettier tables
  survival, # For survival analysis in the lab
  broom.helpers # For tidying model outputs
)
```

```
# Set a consistent theme for our DAGs
theme_set(theme_dag_blank())
```

1. Comprehensive Topic Explanation

1.1 The Central Problem: Association vs. Causation

From our first lecture (and Hernán & Robins (2020), Chapter 1), we know that the fundamental challenge of causal inference is that **association is not causation**.

- **Association** is what we *observe* in the data. It's a comparison of different people in different groups.
  - Association = $\mathbb{E}[Y|A = 1] - \mathbb{E}[Y|A = 0]$
- **Causation** is what we *want to know*. It's a "what if" comparison of the *same* population under two different scenarios.
  - Causation = $\mathbb{E}\left[Y^{a=1}\right] - \mathbb{E}\left[Y^{a=0}\right]$

In the heart transplant example, the association was 0.11 (transplant recipients had a *higher* risk of death), but the causal effect was 0. The difference was **confounding**: sicker patients ($L = 1$) were both more likely to get a transplant ($L \rightarrow A$) and more likely to die ($L \rightarrow Y$).

1.2 The "Gold Standard": The Randomized Controlled Trial (RCT)

The RCT is our conceptual "gold standard" because it provides a clear path to causal inference. By randomly assigning a treatment *strategy* Z, an RCT achieves two critical goals:

1. **Breaks Confounding:** Random assignment Z is, by definition, not caused by any patient characteristic (measured L or unmeasured U). This breaks all backdoor paths between *assignment* and outcome.
2. **Defines a Clear Intervention:** The trial protocol explicitly defines the treatment strategies being compared.

Visually, randomization severs the link between any confounders (L or U) and the intervention, Z.

In the RCT (right), the **Intention-to-Treat (ITT)** effect, which is the causal effect of *assignment* (Z) on the outcome (Y), is unconfounded because the backdoor path $Z \leftarrow U \rightarrow Y$ is broken.

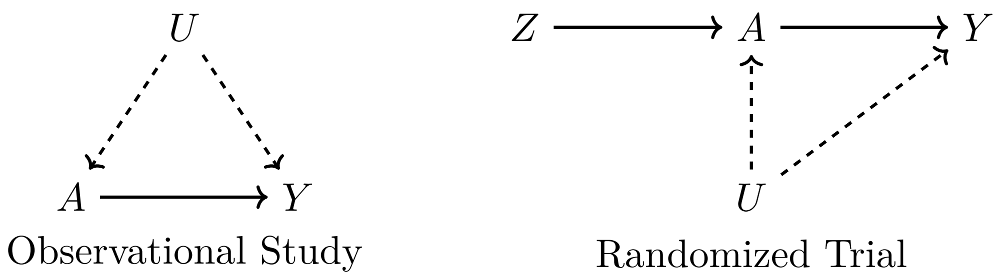

Figure 1: Left (**Observational study**): an unmeasured confounder U (dashed arrows) affects both treatment A and outcome Y, opening the backdoor path A <- U -> Y $A \leftarrow U \rightarrow Y$; therefore a simple contrast $\mathbb{E}[Y|A=1] - \mathbb{E}[Y|A=0]$ is generally biased. Right (**Randomized trial**): assignment Z is randomized and thus not caused by U, so the intention-to-treat (ITT) effect of assignment ($Z \rightarrow Y$) is unconfounded (no backdoor $Z \leftarrow U \rightarrow Y$). *Caveat*: adherence A can still be influenced by U (dashed arrow to A on the right), so the per-protocol contrast comparing $A=1$ vs $A=0$ may remain confounded by U and requires additional adjustment or g-methods to estimate. Dashed arrows indicate unmeasured (latent) variables.

1.3 The Unifying Idea: The Target Trial Framework

Most observational studies are attempts to answer a causal question. The **target trial framework** formalizes this by stating that every observational analysis should be seen as an *emulation* of a hypothetical, pragmatic randomized trial.

If we fail to emulate the trial correctly, we introduce biases *on top of* the usual confounding. By explicitly designing the target trial *first*, we can avoid these "self-inflicted injuries".

1.4 The Seven Protocol Components

To design our observational study "as if" it were a trial, we must explicitly specify the seven components of the target trial's protocol. We will use the example of estimating the effect of aspirin on mortality after colon cancer surgery, as described in Hernán et al. (2016).

1. Eligibility Criteria

- **What it is:** The set of rules that define who is included in the trial.
- **Rationale:** This defines the population to which our causal effect estimate will apply (i.e., its generalizability or transportability). It also ensures that all participants are able to *initiate* either of the treatment strategies being compared.
- **Target Trial Example:** "Individuals with a new diagnosis of colon cancer, no prior aspirin use, and no contraindications, who survived 1 month post-surgery".

- **Observational Emulation Challenge:** We must apply these *same* criteria to our observational data. A common failure is **prevalent user bias**, where we include people *already* on aspirin. This is a form of selection bias, as it excludes those who may have tried aspirin and stopped (perhaps due to side effects) or who died shortly after initiation. The correct approach is a **new-user design**, where we only include individuals at the moment they *first* become eligible to initiate a strategy.

2. Treatment Strategies

- **What it is:** The interventions we wish to compare. These must be precisely defined and sustained over the follow-up.
- **Rationale:** A vague question like "the effect of aspirin" is ill-defined. We must specify *which* aspirin strategy (e.g., dose, timing, duration).
- **Target Trial Example:**
  - **Strategy g_1:** "Initiate daily aspirin (81mg) at T_0 and continue daily, unless a contraindication or toxicity arises."
  - **Strategy g_0:** "Do not initiate aspirin at T_0 and do not start during follow-up."
- **Observational Emulation Challenge:** We must find individuals in our data whose observed actions map to these strategies. Note that g_1 is a **dynamic strategy**: the treatment at time k depends on a time-varying covariate (toxicity).

3. Assignment Procedure

- **What it is:** The method used to assign individuals to a treatment strategy.
- **Rationale:** This is the core of the causal inference.
- **Target Trial Example: Randomization** at T_0. This ensures the groups are exchangeable (comparable) at baseline. $Z \perp\!\!\!\perp (Y^a, L, U)$.
- **Observational Emulation Challenge:** This is the one component we *cannot* replicate. Instead, we must *assume* **conditional exchangeability** (or "no unmeasured confounding"). We assume that, conditional on the *measured* baseline confounders L (age, disease history, etc.), assignment to the aspirin strategy is "as if" random. We then use adjustment methods (matching, weighting, etc.) to emulate randomization.

4. Follow-up

- **What it is:** When follow-up starts and ends.
- **Rationale:** This must be precisely aligned with assignment and eligibility to avoid bias.
- **Target Trial Example:** Follow-up *must* start at **Time Zero** (T_0)—the moment of randomization. It ends at a pre-specified time (e.g., 5 years) or when the outcome occurs (e.g., death).

- **Observational Emulation Challenge:** Our observational follow-up *must also* start at Time Zero—the moment eligibility is met and the treatment strategy is *conceptually* assigned (e.g., at 1-month post-surgery). Any misalignment of T_0 with eligibility and assignment leads to bias (see Section 2.1).

5. Outcome

- **What it is:** What is being measured, and how.
- **Rationale:** Must be defined clearly and ascertained consistently across groups.
- **Target Trial Example:** "All-cause mortality within 5 years".
- **Observational Emulation Challenge:** We must ascertain the outcome (e.g., death from a national registry) in the same way for both groups to avoid differential measurement bias.

6. Causal Contrast (Estimand)

- **What it is:** The specific causal question we are asking.
- **Rationale:** We must decide if we care about the effect of *assignment* or the effect of *adherence*. These are two different causal questions.
- **Target Trial Example:**
  - **Intention-to-Treat (ITT) Effect:** The causal effect of *assignment* to the strategies at T_0, regardless of subsequent adherence.
  - **Per-Protocol (PP) Effect:** The causal effect of *adhering* to the strategies for the entire follow-up.
- **Observational Emulation Challenge:** We must also choose.
  - *ITT-analog:* Effect of *initiation* at T_0.
  - *PP-analog:* Effect of *adherence* over time. This is more complex as it involves time-varying treatments.

7. Analysis Plan

- **What it is:** The statistical method used to estimate the chosen causal contrast.
- **Rationale:** The method must match the causal contrast and assumptions.
- **Target Trial (ITT):** Unadjusted (or baseline-adjusted) comparison of risks/rates (e.g., Kaplan-Meier curves, risk ratio).
- **Observational Emulation (ITT-analog):** An adjusted comparison (e.g., matching, IP weighting) to account for *baseline* confounding.
- **Observational Emulation (PP-analog):** A g-method (e.g., IP weighting of a marginal structural model, g-formula) to account for *time-varying* confounding.

2. Intuitive Clarifications of Challenging Ideas

2.1 The Peril of Misaligned Time Zero (T_0)

The most common "self-inflicted injury" in observational research is failing to align T_0 (the start of follow-up) with eligibility and treatment assignment (Hernán et al., 2016).

Imagine T_0 as the **starting line of a race**.

- **Target Trial (Correct):** An official (randomization) tells everyone who is eligible (e.g., "no injuries") to line up at the starting line (T_0). They are assigned a strategy ("run fast" vs. "run slow"). The official fires a gun, and *everyone* starts the race (follow-up) at that *exact moment*.
- **Observational Emulation (Flawed):** We are watching a marathon that has already started.

⚠ Emulation Failure 1: Prevalent User Bias

The Error: We start our study at the 10-mile mark. Our T_0 is 10 miles into the race. We compare the people already wearing the "fast shoes" (prevalent users) to those in "slow shoes".
The Bias: This is **selection bias**. The only people we see in the "fast shoes" group are those who *survived* the first 10 miles (i.e., didn't get a cramp, quit, or collapse). We have selected a biased sample of "survivors" of the early effects of the shoes. This bias is particularly problematic if the treatment has short-term harms.
The Fix: A **new-user design**. We must stand at the *real* starting line (T_0) and only include people as they *initiate* their strategy.

⚠ Emulation Failure 4: Immortal Time Bias

The Error: We stand at the starting line (T_0). We fire the gun. We wait until the 10-mile mark and *then* we assign people to groups: "Fast Shoe Group" (if they are wearing fast shoes *at 10 miles*) vs. "Slow Shoe Group" (all others).
The Bias: This is a severe, self-inflicted bias. The "Fast Shoe Group" *by definition* had to survive the first 10 miles (the **"immortal time"**) to even be included in their group. This guarantees they will look better than the "Slow Shoe Group," which includes all the people who dropped out at mile 6. This is a form of selection bias where we condition on survival.
The Fix: Align assignment with T_0. Classify people into "Fast Shoe Strategy" vs. "Slow Shoe Strategy" based *only* on the shoes they are wearing at the T_0 starting line (or, more realistically, that they *initiate* wearing at T_0).

2.2 The Two Key Causal Questions: ITT vs. PP

This is a major point of confusion from Hernán & Robins (2020) (Chapter 22). Both are valid causal contrasts, but they answer *different questions* by contrasting *different interventions*.

Let Z be the *assigned* strategy (what the doctor/randomization tells you to do) and $\overline{A}$ be the *received* strategy (what you actually do).

1. The Intention-to-Treat (ITT) Effect

- **Causal Question:** What is the causal effect of *assignment* to a strategy?
- **Intervention:** $Z = 1$ ("Assign to g_1") vs. $Z = 0$ ("Assign to g_0").
- **Causal Contrast:** $\mathbb{E}\left[Y^{z=1}\right] - \mathbb{E}\left[Y^{z=0}\right]$
- **Plain English:** "What is the 5-year mortality risk if we *assign* everyone to the aspirin strategy, versus if we *assign* everyone to the no-aspirin strategy?"
- **Key Feature:** This is a *pragmatic* or *policy-level* question. It answers: "What will happen if we *implement a program* that encourages everyone to take aspirin?" It already accounts for the fact that in the real world, people are not perfectly adherent.
- **Identification in an RCT:** Because Z is randomized, Z is unconfounded. We can estimate this directly: $\mathbb{E}[Y|Z = 1] - \mathbb{E}[Y|Z = 0]$ (assuming no loss to follow-up).

2. The Per-Protocol (PP) Effect

- **Causal Question:** What is the causal effect of *adhering* to a strategy?
- **Intervention:** $\overline{a} = \overline{g}_1$ ("Receive g_1") vs. $\overline{a} = \overline{g}_0$ ("Receive g_0").
- **Causal Contrast:** $\mathbb{E}\left[Y^{\overline{a}=\overline{g}_1}\right] - \mathbb{E}\left[Y^{\overline{a}=\overline{g}_0}\right]$
- **Plain English:** "What is the 5-year mortality risk if everyone *actually took aspirin* as prescribed, versus if everyone *actually took no aspirin* as prescribed?"
- **Key Feature:** This is the effect of the *treatment itself*. It's what a patient who *plans to adhere* wants to know.
- **Identification in an RCT:** This is hard! We *cannot* just compare the people who happened to adhere (the "naïve per-protocol" analysis).
- **Why is the "naïve" analysis biased?** Adherence ($\overline{A}$) is a post-randomization behavior. It is confounded by factors (L_k, U_k) like side effects or new symptoms, which themselves predict the outcome Y. This creates **treatment-confounder feedback** and selection bias.
- **Conclusion:** To estimate the PP effect, we must treat the RCT (or our observational study) as an observational study of a time-varying treatment and adjust for the time-varying confounders that predict adherence. This is why we need the g-methods from Part III.

Table 1: Comparison of the Intention-to-Treat (ITT) Effect vs. the Per-Protocol (PP) Effect.{#tbl-itt-vs-pp}

Feature	Intention-to-Treat (ITT) Effect	Per-Protocol (PP) Effect
Causal Question	What is the effect of *assignment* to a strategy?	What is the effect of *adhering* to a strategy?
Intervention	$Z = 1$ (Assignment) vs. $Z = 0$ (Assignment)	$\bar{a} = \bar{g}_1$ (Adherence) vs. $\bar{a} = \bar{g}_0$ (Adherence)
Causal Contrast	$\mathbb{E}\left[Y^{z=1}\right] - \mathbb{E}\left[Y^{z=0}\right]$	$\mathbb{E}\left[Y^{\bar{a}=\bar{g}_1}\right] - \mathbb{E}\left[Y^{\bar{a}=\bar{g}_0}\right]$
Key Challenge	In RCTs: None (if no loss to follow-up).In Obs: Adjusting for *baseline* confounding.	In RCTs & Obs: Adjusting for *baseline AND time-varying* confounding (non-adherence).
Adjustment	Baseline Adjustment (e.g., regression, matching).	G-methods (e.g., IP weighting, g-formula).

3. Simple, Hand-Calculation Numerical Example(s)

Example 1: Diagnosing Immortal Time Bias (Failure 4)

Let's use a simple example to see the bias in action.

- **Goal:** Does Drug A reduce 1-year mortality?
- **Data:** 100 patients. 50 die at 6 months. 50 survive the full 12 months.
- **Incorrect Design:**
  - T_0 = Study start.
  - "Treated Group" ($A = 1$): Defined as those who *survive* at least 9 months *and* are taking the drug at 9 months.
  - "Untreated Group" ($A = 0$): All others.
- **Analysis:**
  - The 50 patients who died at 6 months *cannot* be in the "Treated Group" because they didn't survive to 9 months. They are all, by definition, in the "Untreated Group".
  - The 50 patients who survived 12 months are the only ones eligible for the "Treated Group". Let's say 25 of them took the drug for >9 months (so they land in $A = 1$) and 25 did not (so they land in $A = 0$).
- **Calculate the Biased Risk:**
  - **Risk (Treated, $A = 1$):**

    - N = 25 (the survivors who took the drug).
    - Deaths = 0 (they all survived 12 months).
    - Risk = 0 / 25 = **0%**
  - **Risk (Untreated, $A = 0$):**
    - N = 75 (the 50 who died early + the 25 survivors who didn't take the drug).
    - Deaths = 50 (the 50 who died early).
    - Risk = 50 / 75 = **66.7%**

- **Biased Conclusion:** The drug appears to be miraculously effective (Risk Ratio = 0). The bias comes from the 9 months of "immortal time" we guaranteed for the treated group.

Example 2: Diagnosing Prevalent User Bias (Failure 1)

- **Goal:** Does Drug A prevent 1-year mortality?
- **Data:** We recruit 100 patients on Jan 1, 2024. This is our T_0.
- **Incorrect Design:**
  - We define our groups based on their drug use *at* T_0.
  - "Treated Group" ($A = 1$): 50 patients who are *already* taking Drug A on Jan 1 (prevalent users).
  - "Untreated Group" ($A = 0$): 50 patients who are *not* taking Drug A on Jan 1.
- **The Unseen Problem:** The drug has a dangerous side effect: 10% of people who initiate it die within the first month. In 2023, 55 people *started* the drug. 5 died, leaving the 50 "healthy survivors" we see in our $A = 1$ group. The 50 people in our $A = 0$ group are just the general population.
- **Calculate the Biased Risk:**
  - Assume the "baseline" 1-year risk for everyone is 20%.
  - **Risk (Treated, $A = 1$):**
    - N = 50. These are "healthy survivors" of the initial risk. Let's say their 1-year risk is now only 10% (5 deaths).
    - Risk = 5 / 50 = **10%**
  - **Risk (Untreated, $A = 0$):**
    - N = 50. These are the general population. Their risk is 20% (10 deaths).
    - Risk = 10 / 50 = **20%**
- **Biased Conclusion:** The drug looks protective (RR = 0.5). We've introduced selection bias by only including the "prevalent users" who were healthy enough to survive the initial high-risk period.

4. R Code Examples & Interpretation

Let's use `dagitty` and `ggdag` to visualize the causal structures for our key concepts.

1. DAG: ITT vs. Naïve Per-Protocol in an RCT

In an RCT, the **ITT effect** ($Z \to \ldots \to Y$) is unconfounded. But the **PP effect** ($A \to Y$) is confounded by U (e.g., frailty, motivation) which affects both adherence (A) and outcome (Y). A "naïve" analysis that just compares $A = 1$ to $A = 0$ will be biased by the open backdoor path $A \leftarrow U \to Y$.

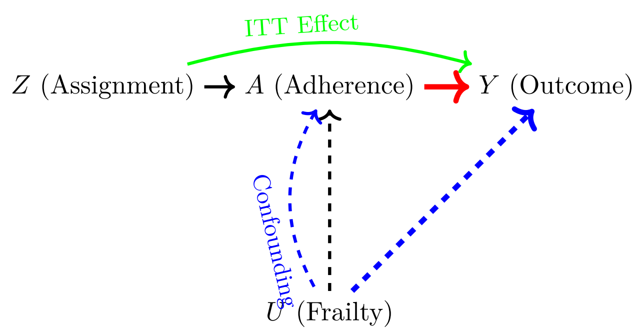

Figure 2: DAG for an RCT. The ITT effect ($Z \to Y$) is unconfounded. The green arrow indicates the ITT effect of assignment (it is not a direct effect of Z on Y, but goes through A). The PP effect ($A \to Y$) is confounded by U. Note that the red arrow indicates the causal effect of interest for the PP analysis, while the blue dashed arrows indicate confounding paths.

2. DAG: Per-Protocol Emulation with Time-Varying Confounding

This is the key challenge for estimating the PP effect. To get the effect of the *full strategy* (A_0, A_1) on Y, we must adjust for L_0 and L_1. But L_1 (e.g., a side effect) is *caused by* A_0.

This creates **treatment-confounder feedback**. As we see, L_1 is:

1. A **confounder** for the $A_1 \to Y$ effect (it opens the backdoor path $A_1 \leftarrow L_1 \to Y$).
2. A **mediator** for the $A_0 \to Y$ effect (it is on the causal path $A_0 \to L_1 \to Y$).

Standard regression adjustment for L_1 would block part of the A_0 effect and (as we'll see in Ch. 20) can introduce selection bias. This is why we need g-methods.

5. Hands-On R Lab

STAT 579: R Lab 2 Emulating a Target Trial: Aspirin & Mortality

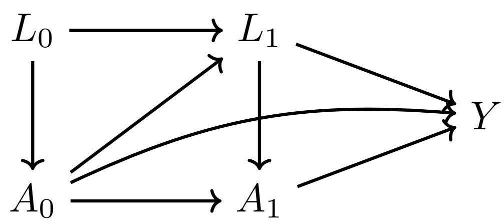

Figure 3: Target trial emulation with treatment-confounder feedback. L_1 is a time-varying confounder for the $A1 \to Y$ effect that is also affected by past treatment A_0.

Lab Objectives

1. Simulate a "messy" observational dataset that includes baseline confounders and time-to-event data.
2. Perform a **biased analysis** using a design that fails to emulate the target trial (Emulation Failure 4: Immortal Time Bias).
3. Perform a **correct emulation** of the target trial by:
   - Defining eligibility at time zero.
   - Assigning treatment strategies at time zero (using a grace period).
   - Starting follow-up at time zero.
4. Adjust for baseline confounding in the emulated trial using a regression model.
5. Compare and interpret the results from the biased and correct analyses.

1. Warm-up & Setup

Reading: Before starting the lab, please review Hernán et al. (2016), sections 2 and 3.

Warm-up Questions (Discuss with a partner):

1. What are the 7 components of the aspirin target trial described in Section 2 of the paper?
2. What is "immortal time"? How does Emulation Failure 4 (Fig 1) guarantee that the treated group will have a "period… during which the risk is guaranteed to be exactly zero"?

First, let's load our packages and set a random seed for reproducibility.

```
pacman::p_load(tidyverse, survival, lubridate, gtsummary)
set.seed(1989) # For reproducibility
```

2. Simulate "Messy" Observational Data

We will simulate a "raw" dataset of 2,000 patients who had colon cancer surgery. This represents the messy electronic medical record (EMR) data we get *before* applying our target trial design.

- `U_frailty` is an **unmeasured** "healthy/frail" status. Healthy ($U = 0$) patients are more likely to take aspirin and less likely to die. This is our unmeasured confounder.
- `age` and `history_chd` are our *measured* confounders (L).
- Aspirin initiation can happen any time after surgery.
- Death can happen any time after surgery.

```r
n <- 2000
dat_raw <- tibble(
  id = 1:n,
  # U: Unmeasured confounder
  U_frailty = rbinom(n, 1, 0.4), # 40% are frail

  # L: Measured confounders
  age = rnorm(n, 60, 8),
  history_chd = rbinom(n, 1, 0.2 + U_frailty * 0.2), # Frail more likely to have CHD

  # Surgery Date (our "anchor" event)
  surgery_date = as.Date("2020-01-01") + runif(n, 0, 365),

  # A: Aspirin Initiation
  # Healthy (U=0) and younger patients are more likely to initiate
  prob_asa = plogis(
    -1 - 0.5 * U_frailty + 0.1 * (age - 60) - 0.2 * history_chd
  ),
  initiates_asa = rbinom(n, 1, prob_asa),
  asa_init_date = if_else(
    initiates_asa == 1,
    surgery_date + runif(n, 1, 730), # Initiates 1 day to 2 yrs post-op
    as.Date(NA)
  ),

  # Y: Death
  # We simulate a "true" survival time.
  # Hazard is higher for frail, older, CHD patients
  log_haz_death = -5 + 1.5 * U_frailty + 0.05 * (age - 60) + 0.5 * history_chd,
```

```
  # We generate a time-to-death assuming no aspirin
  time_to_death_no_asa = rexp(n, exp(log_haz_death)),

  # True Causal Effect: Aspirin has a Hazard Ratio of 0.7
  time_to_death = if_else(
    initiates_asa == 1,
    time_to_death_no_asa / 0.7,
    time_to_death_no_asa
  ),

  death_date = surgery_date + time_to_death
)

# Clean up raw data
dat_raw <- dat_raw |>
  select(
    id,
    U_frailty,
    age,
    history_chd,
    surgery_date,
    asa_init_date,
    death_date
  )
```

3. Task 0: Explore the Raw Data

Before any analysis, look at the raw data. Get a feel for the variables.

```
glimpse(dat_raw)

summary(dat_raw)

# How many people initiated aspirin?
dat_raw |>
  summarise(
    n_total = n(),
```

```
    n_initiators = sum(!is.na(asa_init_date)),
    percent_initiators = 100 * (n_initiators / n_total)
  )
```

4. Task 1: The Biased "Immortal Time" Analysis

Goal: Perform a flawed analysis based on "Emulation Failure 4".

- T_0: 1 month post-surgery (our "anchor").
- **Groups:**
  - `"Ever Aspirin"`: Anyone who *ever* initiated aspirin (even 2 years later).
  - `"Never Aspirin"`: Anyone who never did.
- **Follow-up:** Start at T_0 for 5 years.

```
# STUDENT TO FILL IN
dat_biased <- dat_raw |>
  mutate(
    time_zero = surgery_date + months(1), # 1 month after surgery_date

    # Biased group assignment based on *any* future use
    strategy_group = if_else(
      is.na(asa_init_date),
      "Never Aspirin",
      "Ever Aspirin"
    ),

    # Follow-up ends at 5 years (1826 days) post-T0 or at death
    followup_end = time_zero + days(1826),
    time_to_event = as.numeric(
      pmin(death_date, followup_end, na.rm = TRUE) - time_zero
    ),
    status = if_else(!is.na(death_date) & death_date <= followup_end, 1, 0)
  ) |>
  # Filter out anyone who died *before* T0
  dplyr::filter(time_to_event > 0)
s
# Run the Cox model
cox_biased <- coxph(
```

```r
  Surv(time_to_event, status) ~ strategy_group + age + history_chd,
  data = dat_biased
)

# Display the biased result
tbl_regression(cox_biased, exponentiate = TRUE)
```

❗ Solution: Task 1

```r
# STUDENT TO FILL IN

dat_biased <- dat_raw |>
  mutate(
    time_zero = surgery_date + months(1),

    # Biased group assignment based on *any* future use
    strategy_group = if_else(
      is.na(asa_init_date),
      "Never Aspirin",
      "Ever Aspirin"
    ),

    # Follow-up ends at 5 years (1826 days) post-T0 or at death
    followup_end = time_zero + days(1826),
    time_to_event = as.numeric(
      pmin(death_date, followup_end, na.rm = TRUE) - time_zero
    ),
    status = if_else(!is.na(death_date) & death_date <= followup_end, 1, 0)
  ) |>
  # Filter out anyone who died *before* T0
  dplyr::filter(time_to_event > 0)

# Run the Cox model
cox_biased <- coxph(
  Surv(time_to_event, status) ~ strategy_group + age + history_chd,
  data = dat_biased
)

# Display the biased result
tbl_regression(cox_biased, exponentiate = TRUE)
```

5. Task 2: Emulating the Target Trial (Correctly)

Now, let's follow the target trial protocol. We will align T_0, eligibility, and treatment assignment.

- T_0: 1 month post-surgery.

- **Eligibility:** Must be alive at T_0.
- **Strategies:** We'll use a 3-month (90-day) **grace period** to make the emulation realistic.
  - `Strategy 1 (Aspirin)`: Initiates aspirin between T_0 and $T_0 + 90$ days.
  - `Strategy 0 (No Aspirin)`: Does *not* initiate aspirin between T_0 and $T_0 + 90$ days.
- **Analysis:** ITT-analog. We compare the groups *as defined at baseline* and follow them for 5 years, regardless of what they do after the grace period.

```r
# STUDENT TO FILL IN
dat_tt <- dat_raw |>
  mutate(
    time_zero = `...`,
    followup_end = `...`,

    # Define strategy assignment *at T0* (using a 90-day grace period)
    asa_within_grace = !is.na(asa_init_date) &
      asa_init_date >= `...` &
      asa_init_date <= (`...` + days(90)),

    strategy_group = if_else(
      asa_within_grace == TRUE,
      "Aspirin Strategy",
      "No Aspirin Strategy"
    ),

    # Define outcome and follow-up time *from T0*
    time_to_event = as.numeric(pmin(`...`, `...`, na.rm = TRUE) - `...`),
    status = if_else(!is.na(`...`) & `...` <= `...`, 1, 0)
  ) |>
  # Apply eligibility criteria AT T0
  dplyr::filter(time_to_event > 0) # Must be alive at T0

# Run the Cox model, adjusting for baseline confounders
cox_tt <- coxph(
  Surv(time_to_event, status) ~ strategy_group + age + history_chd,
  data = dat_tt
)

# Display the target trial emulation result
tbl_regression(cox_tt, exponentiate = TRUE)
```

❗ Solution: Task 2

```r
dat_tt <- dat_raw |>
  mutate(
    time_zero = surgery_date + months(1),
    followup_end = time_zero + days(1826),

    # Define strategy assignment *at T0* (using a 90-day grace period)
    asa_within_grace = !is.na(asa_init_date) &
      asa_init_date >= time_zero &
      asa_init_date <= (time_zero + days(90)),

    strategy_group = if_else(
      asa_within_grace == TRUE,
      "Aspirin Strategy",
      "No Aspirin Strategy"
    ),

    # Define outcome and follow-up time *from T0*
    time_to_event = as.numeric(
      pmin(death_date, followup_end, na.rm = TRUE) - time_zero
    ),
    status = if_else(!is.na(death_date) & death_date <= followup_end, 1, 0)
  ) |>
  # Apply eligibility criteria AT T0
  dplyr::filter(time_to_event > 0) # Must be alive at T0

# Run the Cox model, adjusting for baseline confounders
cox_tt <- coxph(
  Surv(time_to_event, status) ~ strategy_group + age + history_chd,
  data = dat_tt
)

# Display the target trial emulation result
tbl_regression(cox_tt, exponentiate = TRUE)
```

(*End of Lab*)

6. Common Pitfalls & Checks

⚠ Common Pitfall: Misaligned Time Zero

The single biggest mistake in observational studies of sustained treatments is misaligning T_0. The Hernán et al. (2016) paper gives 4 classic examples. Always check your (or others') analyses for these:

1. **Failure 1 (Prevalent User Bias):** T_0 is *after* strategy assignment.
   - *Example:* Study of statins and dementia. T_0 is age 65. "Treated" group are those taking statins at age 65.
   - *Bias:* Selection bias. You miss everyone who tried statins and quit (maybe due to side effects) and everyone who died before 65 (maybe due to the statin).
   - *Fix:* A **new-user design**. We must stand at the *real* starting line (T_0) and only include people as they *initiate* their strategy.
2. **Failure 2 (Eligibility after T_0):** T_0 is at eligibility, but *after* strategy assignment.
   - *Example:* Study of aspirin. T_0 is 1-month post-surgery (eligibility). But the "treated" group is defined as those who started aspirin *before* surgery.
   - *Bias:* Selection bias. The "treated" group is selected on surviving *until* T_0.
3. **Failure 3 (Immortal Time before Eligibility):** T_0 is *before* eligibility is fully met.
   - *Example:* Study of post-surgery aspirin. T_0 is the date of *cancer diagnosis*. "Treated" is defined as starting aspirin at diagnosis.
   - *Bias:* Immortal time. The outcome (death) can't happen between T_0 (diagnosis) and the end of eligibility (1-month post-surgery).
4. **Failure 4 (Immortal Time after T_0):** T_0 is at eligibility, but strategy is assigned *later*.
   - *Example:* Our hand-calculation and lab example. T_0 is 1-month post-surgery. "Treated" is defined as "used aspirin for ≥ 6 months."
   - *Bias:* Immortal time. The treated group is guaranteed to be alive and in follow-up for at least 6 months.

Check Your Understanding

1. **Question:** An analyst wants to study the effect of statins on heart attacks. They define the "treated" group as anyone who filled $\geq$ 2 statin prescriptions and the "control" group as anyone who filled 0. They start follow-up for *everyone* at Jan 1, 2020. A patient who dies on Jan 15, 2020 (and filled 0 prescriptions) is put in the control group. What bias is this?

- **Answer:** This is classic immortal time bias (Emulation Failure 4). The "treated" group is guaranteed to be immortal for the time it takes to fill 2 prescriptions, while the "control" group includes people who die early.

2. **Question:** What is the *only* way to correctly estimate a per-protocol effect in a trial with non-adherence and time-varying prognostic factors (e.g., factors that predict both adherence and the outcome)?

- **Answer:** You must use a method that can handle time-varying treatment-confounder feedback, such as the g-methods (IP weighting, g-formula, g-estimation).

3. **Question:** You are emulating a target trial. You find that many people in your observational data don't start their assigned "aspirin" strategy exactly at T_0, but rather 2-3 months later. If you *exclude* them, what bias might you introduce? What is a better approach?

- **Answer:** Excluding them could introduce selection bias if the reasons for their delay are related to the outcome. A better approach is to specify a **grace period** (e.g., 3 months) in your target trial protocol. This defines the "Aspirin Strategy" as "initiating within 3 months". This makes the emulation more realistic and increases the number of eligible individuals.

Appendix: Advanced Theory & Derivations

Formalizing the ITT and Per-Protocol Estimands

Let's formalize the concepts from Section 2. We use the potential outcomes framework from Hernán & Robins (2020) (Chapter 22).

- Let Z_i be the **randomized assignment** for individual i at T_0.
  - $Z = 1$: Assigned to "strategy g_1" (e.g., take aspirin).
  - $Z = 0$: Assigned to "strategy g_0" (e.g., no aspirin).
- Let $\overline{A}_i = (A_{i,0}, A_{i,1}, \dots, A_{i,K})$ be the **received treatment history**.
- Let Y_i be the observed outcome.

The Intention-to-Treat (ITT) Effect

The ITT effect is the causal effect of *assignment*.

- **Potential Outcomes:**
  - $Y_i^{z=1}$: The outcome for individual i *if they had been assigned* to strategy g_1.
  - $Y_i^{z=0}$: The outcome for individual i *if they had been assigned* to strategy g_0.

- **Causal Contrast (Estimand):**

$$ACE_{ITT} = \mathbb{E}\left[Y^{z=1}\right] - \mathbb{E}\left[Y^{z=0}\right]$$

- **Identification:** By randomization, assignment Z_i is independent of the potential outcomes: $(Y_i^{z=1}, Y_i^{z=0}) \perp\!\!\!\perp Z_i$. This is **exchangeability**.
- **Result:** Because of exchangeability, $\mathbb{E}[Y^z] = \mathbb{E}[Y|Z = z]$. Therefore, the causal ITT effect is identified by the simple associational contrast:

$$ACE_{ITT} = \mathbb{E}[Y|Z = 1] - \mathbb{E}[Y|Z = 0]$$

This holds assuming no loss to follow-up (selection bias).

The Per-Protocol (PP) Effect

The PP effect is the causal effect of *adhering* to the strategies.

- **Potential Outcomes:** Let $\overline{g}_1$ and $\overline{g}_0$ be the full treatment *histories* specified in the protocol (e.g., g_1 = "take aspirin daily for 5 years", g_0 = "take no aspirin for 5 years").
  - $Y_i^{\overline{a}=\overline{g}_1}$: The outcome for individual *i if they had perfectly adhered* to strategy g_1.
  - $Y_i^{\overline{a}=\overline{g}_0}$: The outcome for individual *i if they had perfectly adhered* to strategy g_0.
- **Causal Contrast (Estimand):**

$$ACE_{PP} = \mathbb{E}\left[Y^{\overline{a}=\overline{g}_1}\right] - \mathbb{E}\left[Y^{\overline{a}=\overline{g}_0}\right]$$

- **Identification Challenge:** This is *not* identified by a simple associational contrast, even in an RCT.
- **The "Naïve" Analysis (BIAS):** The analysis that compares $\mathbb{E}\left[Y|\overline{A} = \overline{g}_1\right]$ vs. $\mathbb{E}\left[Y|\overline{A} = \overline{g}_0\right]$ is **confounded**.
- **Why?** $\overline{A}$ (received treatment) is a post-randomization behavior. Unmeasured factors U (like frailty, motivation) can affect both *whether* a person adheres ($\overline{A}$) and their outcome (Y). This creates a confounding path: $\overline{A} \leftarrow U \rightarrow Y$.
- **Result:** To estimate the PP effect, we must treat the RCT as an observational study. We must assume **sequential exchangeability** conditional on the measured *time-varying* covariates L_k (e.g., symptoms, side effects) that predict adherence. We must then use g-methods (IP weighting, g-formula) to adjust for this time-varying confounding.

Formalizing "Cloning" for Complex Assignment

In Section 2.2, we discussed two problems: (1) individuals being eligible for multiple strategies at T_0, and (2) grace periods. Both are solved by "cloning" (Hernán & Robins, 2020, Chapter 22).

1. **The Problem:** At T_0, an individual (e.g., $L_0 = 0$) is eligible for both strategy g_1 ("start when $L_1 = 1$") and strategy g_0 ("never start"). Their observed data $\overline{A}$ and $\overline{L}$ are consistent with both.
2. **The Solution:**
   - Create two "clones" of this individual in the dataset.
   - `Clone 1`: Assign to strategy g_1.
   - `Clone 2`: Assign to strategy g_0.
   - Follow both clones over time.
   - If the real person's L_1 becomes 1 and they *start* treatment ($A_1 = 1$), their data are no longer consistent with strategy g_0. We therefore **censor** `Clone 2` at time $k = 1$. `Clone 1` continues.
   - If the real person's L_1 becomes 1 and they *do not* start treatment ($A_1 = 0$), their data are no longer consistent with strategy g_1. We therefore **censor** `Clone 1` at time $k = 1$. `Clone 2` continues.
3. **The Analysis:** This "cloning and censoring" procedure creates a new dataset where assignment is clear at T_0. However, it introduces *time-varying censoring* that is highly informative (it depends on L_k and A_k).
4. **The Result:** We *must* use a g-method (like IP weighting) to handle this induced selection bias. We would create weights SW^C to account for this censoring, as described in Chapter 21 of Hernán & Robins (2020).

This advanced technique is the formal basis for emulating target trials with complex, dynamic strategies or grace periods.

References

Hernán, M. Á., & Robins, J. M. (2020). *Causal inference: What if*. Chapman & Hall/CRC.

Hernán, M. Á., Sauer, B. C., Hernández-Díaz, S., Platt, R., & Shrier, I. (2016). Specifying a target trial prevents immortal time bias and other self-inflicted injuries in observational analyses. *Journal of Clinical Epidemiology*, *79*, 70–75. https://doi.org/10.1016/j.jclinepi.2016.04.014